

1 Introduction

Transmission lines at microwave frequencies exhibit distributed behaviour – their physical length is comparable to the signal wavelength – making simple lumped approximations insufficient unless carefully discretised. **The equivalent circuit model:** A cascaded π -section RLGC ladder network approximates a distributed line and can be simulated in any SPICE-based tool, reducing reliance on expensive EDA solvers. In this report, a microstrip line ($\ell = 99.8$ mm, $W = 200$ μ m, $h = 76$ μ m, $\epsilon_r = 2.94$, $\tan \delta = 0.0016$, copper conductor) is modelled using 48 π -sections valid up to 6 GHz, and analysed in both frequency and time domains.

2 Results – Lossless Line

2.1 Model Design and Lumped Element Calculation

The microstrip effective permittivity ($W/h = 2.63 \geq 1$) and characteristic impedance are:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \cdot \frac{1}{\sqrt{1 + 12h/W}} = 2.381, \quad Z_0 = \frac{120\pi}{\sqrt{\epsilon_{\text{eff}}} \left[\frac{W}{h} + 1.393 + 0.667 \ln \left(\frac{W}{h} + 1.444 \right) \right]} = 49.24 \, \Omega \quad (1)$$

The π -model validity condition requires each section length $\Delta l < \lambda/10$ at f_{max} , giving $N_{\text{min}} = 31$; 48 sections are used for additional accuracy ($\Delta l = 2.08$ mm). Per-unit-length and per-section element values are:

$$L' = \frac{Z_0 \sqrt{\epsilon_{\text{eff}}}}{c_0}, \quad C' = \frac{\sqrt{\epsilon_{\text{eff}}}}{Z_0 c_0}, \quad L_{\text{sec}} = L' \cdot \Delta l = 0.5617 \, \text{nH}, \quad C_{\text{sec}} = C' \cdot \Delta l = 0.2317 \, \text{pF} \quad (2)$$

Verification: $\sqrt{L_{\text{sec}}/C_{\text{sec}}} = 49.24 \, \Omega$. The analytical time of flight is:

$$\tau = \frac{\ell_{\text{total}} \sqrt{\epsilon_{\text{eff}}}}{c_0} = 0.513 \, \text{ns} \quad (3)$$

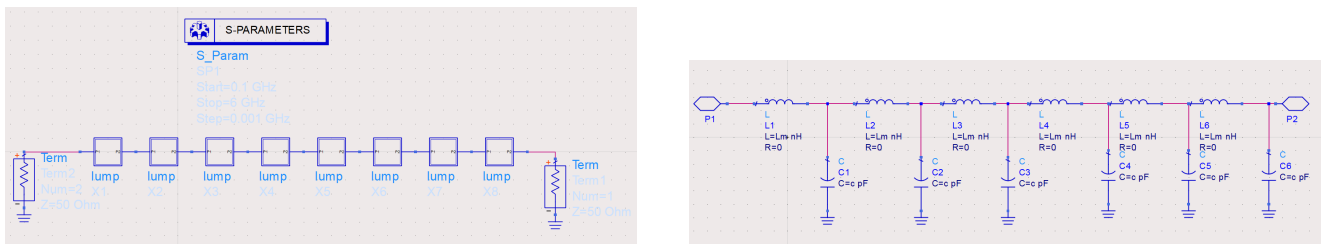


Figure 1: **S-parameter setup in ADS and circuit** (start=0.1GHz, stop=6GHz, step=0.001GHz)(a) Top-level hierarchical schematic of the 48-section lossless microstrip model. (b) Internal π -section: series inductor $L_{\text{sec}} = 0.5617$ nH with shunt capacitors $C_{\text{sec}}/2 = 0.1158$ pF at each port.

2.2 Frequency-Domain S-Parameter Results

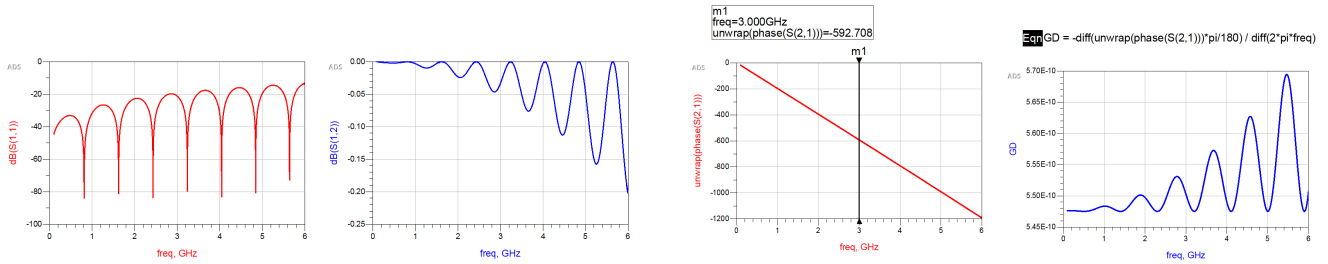


Figure 2: (a) Simulated $|S_{11}|$ and $|S_{21}|$ of the lossless model (0–6 GHz). $|S_{21}|$ remains within -0.20 dB, confirming near-lossless transmission; periodic $|S_{11}|$ nulls correspond to $\lambda/2$ resonances of the ladder. (b) Unwrapped phase of S_{21} (linear, reaching -1200° at 6 GHz) and computed group delay (≈ 0.548 ns, flat below 3 GHz), consistent with the analytical $\tau = 0.513$ ns.

Note: Agreement between your loss-less equivalent circuit model and the s-parameters response (phase and amplitude) has been shown in section 4

2.3 Transient Analysis

The model was excited with a single voltage pulse (width = 1 ns, $V_{\text{high}} = 1$ V) with source and load terminated in $50\ \Omega$. Rise/fall time was varied from 1 ns to 0.001 ns to observe bandwidth-dependent distortion.

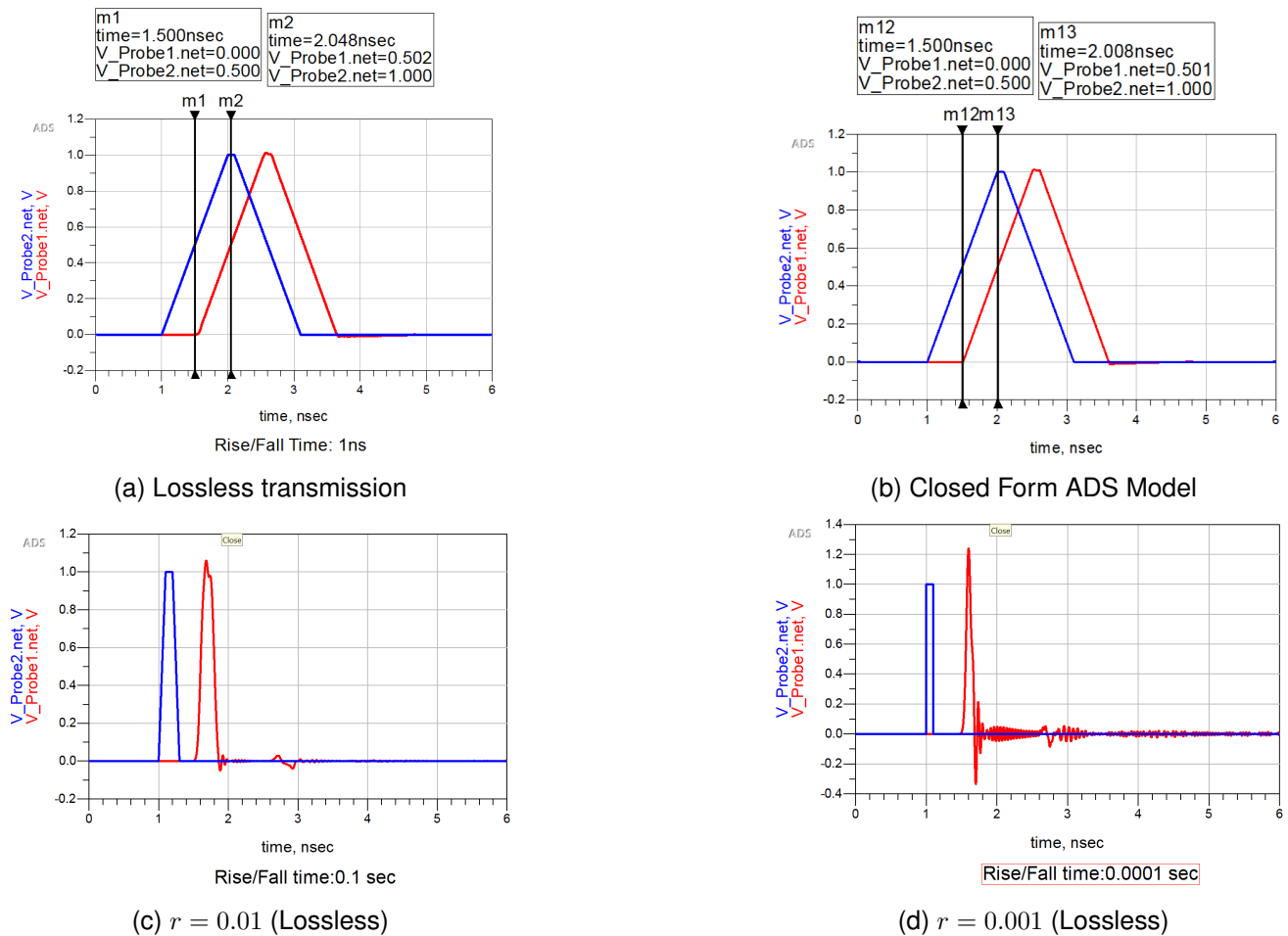


Figure 3: Transient voltage at input (blue) and output (red). (a) Rise time = 1 ns: output is a faithful delayed replica; measured delay = 0.548 ns, agreeing with $\tau = 0.513$ ns. (b) Rise time = 0.001 ns: severe distortion with ringing and overshoot (> 1.2 V), since pulse bandwidth ($\sim 1/t_r \approx 1$ THz) far exceeds the model's 6 GHz validity, exciting LC ladder resonances.

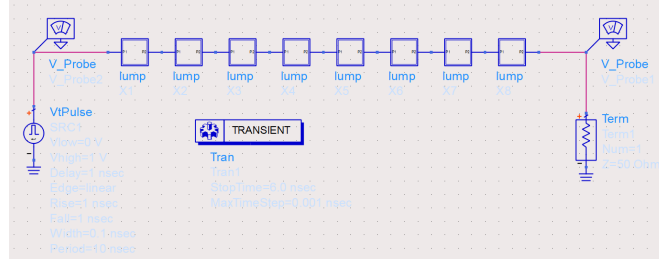


Figure 4: **Transient Simulation Setup** (rise/fall=1 ns, period=10 ns, stoptime=6 ns)

3 Results – Lossy Line

3.1 Loss Mechanisms and Model Parameters

Two loss mechanisms are incorporated. **Conductor loss** (skin effect): current crowds into a surface layer of depth δ , increasing series resistance as $R' \propto \sqrt{f}$:

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} = 0.853 \text{ } \mu\text{m at 6 GHz}, \quad R' = \frac{1}{\sigma \delta W} = 101.04 \text{ } \Omega/\text{m}, \quad R_{\text{sec}} = R' \Delta l = 0.210 \text{ m}\Omega \quad (4)$$

Dielectric loss: molecular polarisation in the substrate dissipates energy, modelled as a shunt conductance $G' \propto f$:

$$G' = \omega C' \tan \delta = 6.30 \text{ mS/m}, \quad G_{\text{sec}} = G' \Delta l = 0.0131 \text{ mS}, \quad R_{\text{par}} = 1/G_{\text{sec}} = 76,334 \text{ } \Omega \quad (5)$$

R_{sec} is placed in series with each inductor; R_{par} in parallel with each capacitor.

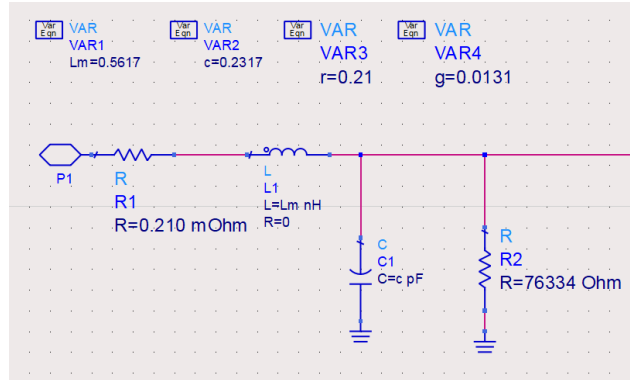


Figure 5: Same top-level hierarchical schematic of the 48-section lossless microstrip model. This figure shows the internal π -section: series inductor $L_{\text{sec}} = 0.5617 \text{ nH}$ with shunt capacitors $C_{\text{sec}}/2 = 0.1158 \text{ pF}$ at each port.

3.2 Lossy Model S-Parameter Results

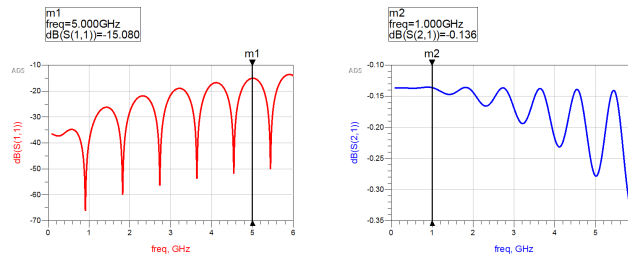


Figure 6: (a) $|S_{21}|$ of the lossy model (0–6 GHz): insertion loss increases from -0.15 dB at DC to -0.35 dB at 6 GHz, reflecting combined $\sqrt{f}$ conductor and linear dielectric loss dependence. Lossy $|S_{21}|$; the $\approx 0.14 \text{ dB}$ baseline offset is the total insertion loss of the line.

3.3 Transient Analysis

The transient response of the lossy model is visually near-identical to the lossless case, which is physically consistent. The total series resistance of the lossy model is $R_{\text{total}} = R_{\text{sec}} \times N = 0.210 \text{ m}\Omega \times 48 = 10.08 \text{ m}\Omega$, which is negligible compared to the 50Ω port terminations. Similarly, the parallel loss resistance $R_{\text{par}} = 76,334 \Omega$ introduces minimal shunt loading. Consequently, the pulse amplitude attenuation is below the simulation resolution, and the damping of LC ringing is imperceptible in the time domain. The effect of losses is more pronounced in the frequency domain, where the cumulative insertion loss across all 48 sections produces a measurable -0.15 dB baseline offset in $|S_{21}|$ and an increasing loss of up to -0.35 dB at 6 GHz . ***This demonstrates that for a short copper microstrip at sub-6 GHz frequencies, conductor and dielectric losses are small and their impact is primarily observable in the frequency domain rather than the time domain.***

4 Analysis and Discussion

4.1 Time Domain and Frequency Domain- Lossless Lines

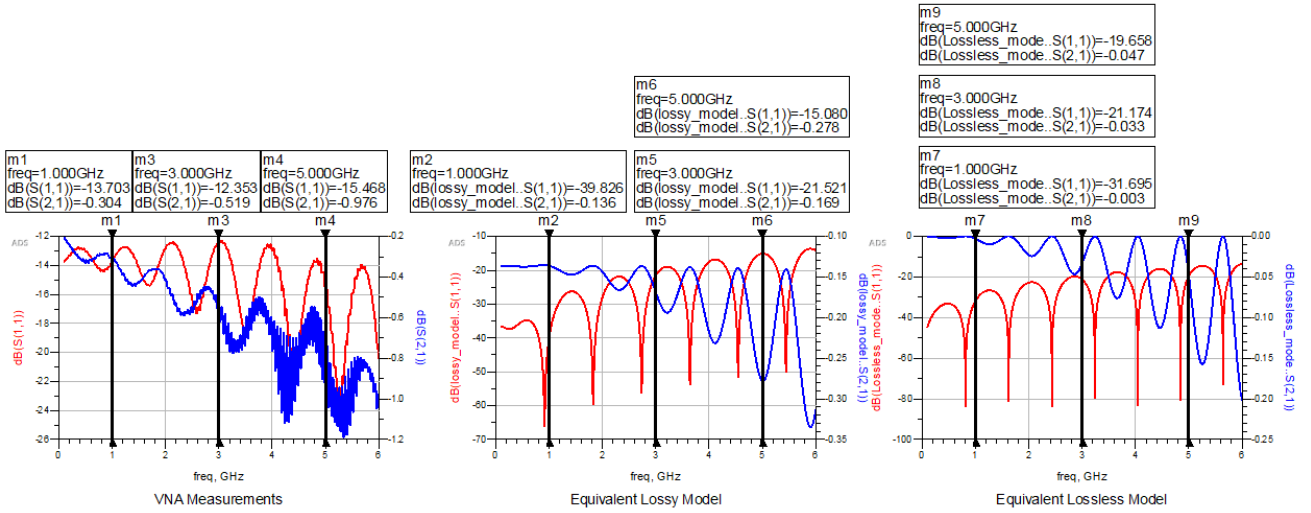


Figure 7: S parameter comparison of all the models

The simulated S-parameter response of the lossless lumped model shows good agreement with the ADS closed-form MLIN reference below 3 GHz , where the π -section approximation is most accurate. The unwrapped phase of S_{21} is linear across the full band, confirming constant group delay consistent with a non-dispersive transmission line. The simulated group delay of $\approx 0.548 \text{ ns}$ agrees with the analytical value of $\tau = 0.513 \text{ ns}$ to within 6.8% , attributable to the finite discretisation of the distributed line into 48 sections. Above 4 GHz , the lumped model exhibits increasing phase deviation from the MLIN reference as the section length Δl approaches $\lambda/10$, introducing parasitic resonances in the LC ladder. The $|S_{21}|$ remains within -0.20 dB across the full band, confirming that amplitude accuracy is well maintained up to 6 GHz .

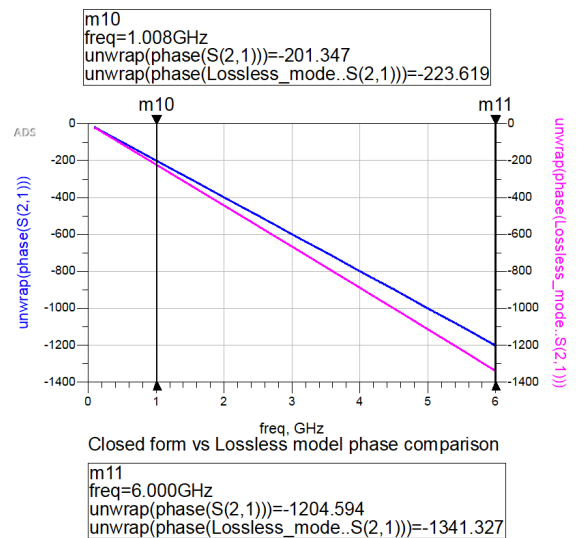


Figure 8: Comparison of unwrapped phase of S_{21} between the lumped π -section model and the ADS MLIN reference model ($0\text{--}6 \text{ GHz}$).

4.2 Simulated and Modelled Response of the Line

Comparing the lumped model against VNA measurements (Figure 7), the lossless model overestimates $|S_{21}|$ relative to measurements since it ignores all losses, while the lossy model shows closer agreement. At 1 GHz, the measured $|S_{21}| = -0.304$ dB compared to the lossy model's -0.136 dB – a discrepancy of 0.168 dB. At 5 GHz this grows to $|S_{21}|_{\text{meas}} = -0.976$ dB vs -0.278 dB (lossy), a discrepancy of 0.698 dB. These inaccuracies arise from three sources. First, the lumped model uses frequency-independent R' and G' values computed at $f_{\text{max}} = 6$ GHz, whereas conductor loss scales as $\sqrt{f}$ and dielectric loss scales linearly with f – both should be recomputed at each frequency for

accuracy. Second, the π -section model becomes increasingly dispersive above 4 GHz as Δl approaches $\lambda/10$, introducing phase errors not present in the physical line. Third, the model does not account for radiation loss, surface wave excitation, or connector parasitics present in the VNA measurement, all of which contribute additional loss at higher frequencies.

4.3 Numerical Parameters of a mmWave TX Line

Extending the model to 50 GHz requires significant changes to maintain validity. The $\lambda/10$ criterion at 50 GHz gives a maximum section length of:

$$\Delta l < \frac{v_p}{10 \times 50 \times 10^9} = \frac{1.944 \times 10^8}{5 \times 10^{10}} = 3.89 \text{ mm} \implies N_{\min} = \left\lceil \frac{99.8}{3.89} \right\rceil \times \frac{50}{6} \approx 430 \text{ sections} \quad (6)$$

The loss parameters also change substantially at 50 GHz. The skin depth reduces to $\delta = 0.296 \mu\text{m}$, increasing R' by $\sqrt{50/6} \approx 2.89\times$ relative to 6 GHz. The dielectric conductance G' increases by $50/6 \approx 8.3\times$. At mmWave frequencies, the PCB substrate itself limits operation: the substrate height $h = 76 \mu\text{m}$ supports surface wave modes above $f_c = c_0/(4h\sqrt{\epsilon_r - 1}) \approx 27$ GHz, causing power leakage into the substrate. Additionally, conductor roughness becomes comparable to δ at 50 GHz, further increasing R' beyond the smooth-conductor approximation. These effects mean the lumped RLGC model becomes inadequate above ~ 30 GHz, and a full-wave EM simulation would be required for accurate mmWave modelling.

5 Conclusion

A lossless and lossy lumped π -section model of a microstrip transmission line were successfully designed and simulated in ADS up to 6 GHz using 48 sections. The lossless model accurately reproduced the expected S-parameter behaviour: $|S_{21}|$ within -0.20 dB, linear unwrapped phase, and a group delay of 0.548 ns consistent with the analytical value of 0.513 ns. Transient simulations confirmed a propagation delay matching the time of flight, with severe distortion at fast rise times demonstrating the model's bandwidth limitation. The lossy model, incorporating skin-effect resistance and dielectric conductance, produced a physically consistent 0.15 dB increase in insertion loss with negligible impact on transient response, confirming that losses are small at sub-6 GHz for this line length. Comparison with VNA measurements revealed discrepancies up to 0.7 dB at 5 GHz, attributed to frequency-independent loss parameters, lumped model dispersion, and unmodelled parasitics. The lumped model offers a

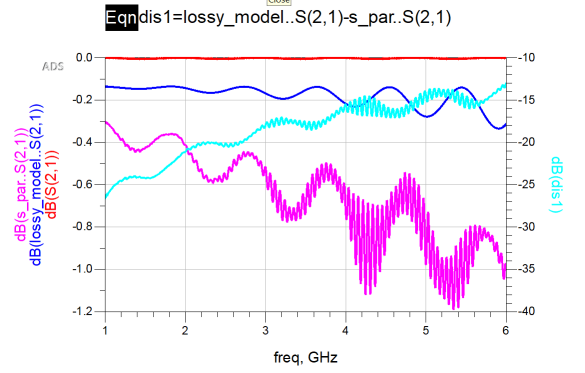


Figure 9: Discrepancy in S_{21} between the lumped model and measured S parameters through a VNA. The ideal microstrip behaviour (red) shows almost 0 insertion loss.

computationally efficient approximation suitable for sub-6 GHz design, but fails at mmWave frequencies where a minimum of ~ 430 sections, frequency-dependent RLGC parameters, and full-wave EM validation are necessary.

Table 1: Lumped element parameters for lossless and lossy π -section models ($\ell = 99.8$ mm, $\epsilon_{\text{eff}} = 2.381$, $Z_0 = 49.24 \Omega$, $N = 48$, $\Delta l = 2.08$ mm, copper, $f_{\text{max}} = 6$ GHz).

Parameter	Symbol	Lossless	Lossy	Unit
Series inductance (p.u.l.)	L'	253.27	253.27	nH/m
Shunt capacitance (p.u.l.)	C'	104.46	104.46	pF/m
Series resistance (p.u.l.)	R'	0	101.04	Ω/m
Shunt conductance (p.u.l.)	G'	0	6.30	mS/m
L per section	L_{sec}	0.5617	0.5617	nH
C per section	C_{sec}	0.2317	0.2317	pF
R per section	R_{sec}	0	0.210	m Ω
Parallel resistance	R_{par}	∞	76,334	Ω
Skin depth (at 6 GHz)	δ	—	0.853	μm
Time of flight	τ	0.513		ns
Group delay (simulated)	τ_g	≈ 0.548		ns
$ S_{21} $ at DC		0	-0.15	dB
$ S_{21} $ at 6 GHz		-0.20	-0.35	dB

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